

I am preparing for my semester exam please provide answers in human written the answers is not long and not short, Answers only medium, subject related to Cyber law & Security and Answers in easy word. For 5-6 marks .

Sure. Send your questions one by one (or multiple together).

I will provide:

- Medium-length answers for **5–6 marks**
- Easy and human-written language
- Proper headings and points
- Cyber Law & Security subject-focused content
- Exam-friendly answers

You can send the next question now.



Q.1 Explain the concept of intrusion detection Systems(IDS). Discuss their working principle, and how they help in protecting networks from attacks.

Intrusion Detection System (IDS)

An Intrusion Detection System (IDS) is a security tool used to monitor network traffic and computer systems for suspicious activities or unauthorized access. It helps in identifying cyber attacks, malware, and security policy violations. IDS acts like a security alarm system for a network.

Working Principle of IDS

The working of an IDS involves the following steps:

1. Monitoring Network Traffic:

IDS continuously observes incoming and outgoing network packets and system activities.

2. Analyzing Data:

It compares the collected data with known attack patterns or unusual behavior.

3. Detection of Threats:

If any suspicious activity is found, the IDS identifies it as a possible intrusion or attack.

4. Generating Alerts:

After detecting a threat, IDS sends warning messages or alerts to the system administrator for immediate action.

Types of IDS**1. Network-based IDS (NIDS):**

Monitors the traffic of the entire network.

2. Host-based IDS (HIDS):

Monitors activities of a single computer or device.

How IDS Protects Networks from Attacks

- Detects hacking attempts and unauthorized access.
- Identifies malware, viruses, and suspicious programs.
- Helps in preventing data theft and network misuse.
- Provides early warning about security threats.
- Assists administrators in improving network security.

Conclusion

Intrusion Detection Systems play an important role in cybersecurity by continuously monitoring systems and detecting possible attacks. They help organizations protect their networks, data, and systems from cyber threats and unauthorized activities.



Q.2 Describe the role of intrusion detection system(IDS) and intrusion prevention system(IPS) in protecting networks. For 5 marks

Role of Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) in Network Protection

Intrusion Detection System (IDS) and Intrusion Prevention System (IPS) are important security tools used to protect computer networks from cyber attacks, unauthorized access, and malicious activities.

Intrusion Detection System (IDS)

An IDS monitors network traffic and system activities to detect suspicious behavior or attacks. It acts like a security alarm system.

Role of IDS:

- Monitors incoming and outgoing network traffic.
- Detects hacking attempts, malware, and unusual activities.
- Generates alerts and warning messages to administrators.
- Helps in identifying security weaknesses and threats.

However, IDS only detects attacks; it does not stop them automatically.

Intrusion Prevention System (IPS)

An IPS not only detects threats but also takes immediate action to stop them. It is placed in-line within the network to block harmful traffic.

Role of IPS:

- Detects and prevents cyber attacks in real time.
- Blocks malicious packets and unauthorized users.
- Prevents malware, viruses, and denial-of-service attacks.
- Improves overall network security by taking automatic actions.

Difference Between IDS and IPS

IDS	IPS
Detects attacks	Detects and blocks attacks
Sends alerts	Takes automatic preventive action
Passive security system	Active security system

Conclusion

IDS and IPS work together to provide strong network security. IDS helps in detecting threats, while IPS prevents attacks before they can damage the network. Both are essential for protecting systems and sensitive data from cyber threats.



Q.3 What is Access control? Discuss the role of biometric systems and the factors influencing their selection. For 5 marks

Access Control

Access control is a security technique used to restrict and manage access to systems, networks, data, and resources. It ensures that only authorized users can access specific information or perform certain actions. Access control helps in protecting sensitive data from unauthorized use.

Role of Biometric Systems in Access Control

Biometric systems use unique physical or behavioral characteristics of a person for identification and authentication. These characteristics include fingerprints, face recognition, iris scan, voice recognition, and hand geometry.

Role of Biometric Systems:

- Provides strong and accurate security.
- Prevents unauthorized access.
- Eliminates the need for passwords or ID cards.
- Improves user authentication and identity verification.
- Widely used in banks, offices, mobile phones, and secure organizations.

Factors Influencing the Selection of Biometric Systems

1. Accuracy:

The system should correctly identify authorized users and avoid false results.

2. Cost:

The biometric device and maintenance cost should be affordable.

3. Security Level:

The system must provide high protection against fraud and hacking.

4. Speed and Performance:

Authentication should be fast and efficient.

5. User Convenience:

The system should be easy to use and comfortable for users.

6. Reliability:

It should work properly under different conditions and environments.

Conclusion

Access control is an essential part of information security. Biometric systems improve access control by providing secure and reliable user authentication. Proper selection of biometric systems depends on factors like accuracy, cost, security, and user convenience.



Q.4 Discuss the major provisions of the Indian IT act, 2000 related to cyber offences. For 5 marks only

Major Provisions of the Indian IT Act, 2000 Related to Cyber Offences

The Information Technology Act, 2000 is the main cyber law in India. It was introduced to provide legal recognition to electronic records, digital signatures, and to control cyber crimes and electronic commerce.

Major Provisions Related to Cyber Offences

1. Hacking and Unauthorized Access (Section 66):

Any person who illegally accesses a computer system, steals data, or damages information is punished under this section.

2. Identity Theft (Section 66C):

Using another person's password, digital signature, or identity information without permission is a punishable offence.

3. Cheating by Personation (Section 66D):

Online frauds and cheating through fake websites, emails, or social media are covered under this section.

4. Publishing Obscene Material (Section 67):

Publishing or transmitting obscene or harmful content electronically is punishable by law.

5. Cyber Terrorism (Section 66F):

Activities that threaten national security through computer systems or networks are treated as cyber terrorism.

6. Protection of Data and Privacy (Section 43A):

Companies handling sensitive personal data must maintain proper security measures to protect information.

7. Digital Signatures and Electronic Records:

The Act gives legal recognition to digital signatures and electronic documents for secure online transactions.

Conclusion

The IT Act, 2000 plays an important role in controlling cyber crimes and ensuring cybersecurity in India. It provides legal protection against various cyber offences and supports safe electronic communication and online transactions.



Q.5 Discuss the ethical issues involved in intellectual Property rights (IPR), including copyright and patents. For 5 marks

Ethical Issues Involved in Intellectual Property Rights (IPR)

Intellectual Property Rights (IPR) are legal rights given to creators and inventors to protect their creations, inventions, and original works. Copyrights and patents are important forms of IPR. Ethical issues arise when these rights are misused or violated.

Ethical Issues Related to Copyright

1. Software and Content Piracy:

Copying software, movies, music, or books without permission is unethical and causes financial loss to creators.

2. Plagiarism:

Using another person's work or ideas without giving proper credit is considered unethical.

3. Unauthorized Sharing:

Sharing copyrighted digital content illegally through the internet violates the rights of the owner.

Ethical Issues Related to Patents

1. Monopoly of Inventions:

Patent holders may charge high prices for products, especially medicines, making them unaffordable for poor people.

2. Blocking Innovation:

Sometimes patents prevent others from improving or developing similar technologies.

3. Patent Misuse:

Some organizations take patents only to control the market instead of encouraging innovation.

Other Ethical Concerns

- Balancing public interest and creator rights.
- Fair use of information and technology.
- Protection of digital content in the online environment.

Conclusion

IPR helps in encouraging creativity and innovation, but ethical issues arise when copyrights and patents are misused or violated. Proper balance between protection of creators and public access is necessary for fair and ethical use of intellectual property.



Q.6 Discuss the physical security of IT assets and backup security measures. Explain their importance in Protecting organizational data and systems. For 5 marks only

Physical Security of IT Assets and Backup Security Measures

Physical security of IT assets means protecting computer systems, servers, networking devices, and data centers from physical damage, theft, and unauthorized access. Backup security measures are used to create and protect copies of important data for recovery during system failure or cyber attacks.

Physical Security of IT Assets

Physical security includes the following measures:

- Use of locks, security guards, and CCTV cameras.
- Restricted access to server rooms and data centers.
- Fire alarms and fire suppression systems.
- Protection against power failure using UPS and generators.
- Proper environmental control such as cooling systems.

Backup Security Measures

Backup security involves storing copies of important data safely.

Common Backup Measures:

- Regular data backups.
- Cloud backup and external storage devices.
- Password protection and encryption of backup files.
- Keeping backup copies at separate secure locations.
- Periodic testing of backup recovery systems.

Importance in Protecting Organizational Data and Systems

- Prevents data loss due to hardware failure, theft, fire, or cyber attacks.
- Ensures business continuity during emergencies.
- Helps in quick recovery of important information.

- Protects sensitive organizational data from unauthorized access.
- Reduces financial and operational losses.

Conclusion

Physical security and backup security measures are essential for protecting organizational IT assets and data. They help organizations maintain data safety, system reliability, and smooth business operations during unexpected situations.



Q.7 What is cyber law? Explain the major issues in E-Business management and Domain Name disputes with suitable examples. For 5 marks only

Cyber Law

Cyber law refers to the laws and legal rules related to the use of computers, internet, digital communication, and online activities. It helps in controlling cyber crimes, protecting digital data, and ensuring safe electronic transactions.

Major Issues in E-Business Management

E-business management involves conducting business activities through the internet. Some major issues are:

1. Security Issues:

Online businesses face risks like hacking, data theft, and cyber attacks.

2. Privacy Issues:

Customer personal and financial information must be protected from misuse.

3. Authentication Problems:

It is difficult to verify the identity of users and online transactions.

4. Payment Fraud:

Fake online payments, phishing, and credit card frauds are common problems.

5. Legal and Regulatory Issues:

Different countries have different cyber laws and business regulations.

Example:

An online shopping website may suffer customer data theft due to weak security systems.

Domain Name Disputes

A domain name dispute occurs when two parties claim rights over the same website domain name. It usually happens when someone registers a famous company or brand name illegally.

Example:

If a person registers a domain like "amaz0n.com" to confuse users and earn profit, it becomes a domain name dispute and may violate trademark laws.

Conclusion

Cyber law is important for regulating online activities and preventing cyber crimes. Proper management of e-business issues and domain name disputes helps in maintaining trust, security, and fairness in the digital world.



Q.8 Explain the different types of cyber crimes with suitable examples. Examine the role of investigation and ethics in addressing cyber security challenges and preventing cyber crimes. For 5 marks

Types of Cyber Crimes and Role of Investigation and Ethics

Cyber crimes are illegal activities performed using computers, networks, or the internet. These crimes target individuals, organizations, and governments.

Different Types of Cyber Crimes

1. Hacking:

Unauthorized access to computer systems or networks.

Example: A hacker stealing data from a company server.

2. Phishing:

Sending fake emails or messages to steal passwords or banking details.

Example: Fake bank email asking for OTP and account information.

3. Identity Theft:

Misusing another person's personal or financial information.

Example: Using someone's debit card details for online shopping.

4. Cyber Stalking:

Harassing or threatening a person through online platforms.

Example: Sending abusive messages repeatedly on social media.

5. Malware Attacks:

Spreading viruses, worms, or ransomware to damage systems.

Example: Ransomware locking a user's files and demanding money.

Role of Investigation in Cyber Security

- Helps in identifying cyber criminals and collecting digital evidence.
- Tracks the source of cyber attacks using forensic tools.
- Supports legal action against offenders.
- Helps organizations improve security after cyber incidents.

Role of Ethics in Preventing Cyber Crimes

- Encourages responsible and legal use of technology.
- Promotes honesty, privacy protection, and respect for digital data.
- Reduces misuse of computer systems and online resources.
- Creates awareness about cyber security and safe internet practices.

Conclusion

Cyber crimes are increasing rapidly with the growth of technology. Proper investigation techniques and ethical practices play an important role in controlling cyber crimes and improving cyber security in society.



Q.9 Analyze the challenges of protecting Personal data in an era of increasing connectivity. For 5 marks

Challenges of Protecting Personal Data in an Era of Increasing Connectivity

In today's digital world, people use the internet, smartphones, cloud services, and social media regularly. Large amounts of personal data are shared online, which creates many security and privacy challenges.

Major Challenges in Protecting Personal Data

1. Increase in Cyber Attacks:

Hackers use phishing, malware, and ransomware attacks to steal personal and financial information.

2. Data Breaches:

Organizations store huge amounts of user data, and weak security systems may lead

to leakage of sensitive information.

3. Excessive Use of Social Media:

People often share personal details online, which can be misused by cyber criminals.

4. Lack of User Awareness:

Many users are unaware of safe online practices and become victims of fraud and identity theft.

5. Cloud and IoT Security Risks:

Connected devices and cloud storage increase the chances of unauthorized access and data misuse.

6. Weak Passwords and Authentication:

Poor password practices make accounts easy targets for attackers.

Importance of Protecting Personal Data

- Maintains privacy and confidentiality.
- Prevents identity theft and financial fraud.
- Builds trust in digital services and online transactions.
- Protects users from cyber crimes and misuse of information.

Conclusion

Protecting personal data has become a major challenge due to increasing internet connectivity and digital dependence. Strong security measures, user awareness, encryption, and proper cyber laws are necessary to ensure data privacy and security.



Q.10 Describe the legal, economic and social challenges associated with information security. For 5 marks

Legal, Economic and Social Challenges Associated with Information Security

Information security is the protection of data and computer systems from unauthorized access, misuse, and cyber attacks. Organizations face many legal, economic, and social challenges in maintaining information security.

Legal Challenges

- Different countries have different cyber laws and regulations.
- Protecting user privacy and personal data is legally important.
- Cyber crimes like hacking, identity theft, and online fraud create legal problems.

- Organizations must follow laws related to data protection and digital transactions.

Economic Challenges

- Implementing security systems and technologies is costly.
- Cyber attacks can cause financial loss, data loss, and business interruption.
- Companies may lose customer trust and market reputation after security breaches.
- Regular maintenance, updates, and employee training increase expenses.

Social Challenges

- Lack of awareness about cyber security among users.
- Misuse of social media and online platforms for fraud and cyber bullying.
- Privacy concerns due to excessive sharing of personal information online.
- Ethical issues related to data misuse and unauthorized surveillance.

Conclusion

Information security is essential in the digital age, but organizations face many legal, economic, and social challenges in protecting data and systems. Proper security policies, user awareness, and strong cyber laws are necessary to overcome these challenges.



Q.11 Differentiate between software privacy and plagiarism with suitable examples from digital media. For 5 marks

Difference Between Software Privacy and Plagiarism

Software privacy and plagiarism are two different concepts related to information technology and digital media. Software privacy focuses on protecting user data, while plagiarism refers to copying someone else’s work without permission or credit.

Basis	Software Privacy	Plagiarism	
Meaning	Protection of personal and sensitive data in software systems	Copying another person’s work or ideas without proper acknowledgment	
Purpose	To maintain confidentiality and security of user information	To protect originality and intellectual property	
Related To	User data and information security	Copyright and ethical use of content	

Basis	Software Privacy	Plagiarism
Example	A mobile app protecting user passwords and banking details	Copying an article, image, or code from the internet and presenting it as own work
Impact	Data theft, privacy violation, and misuse of information	Copyright violation and loss of originality

Examples from Digital Media

- **Software Privacy Example:**
Social media applications like messaging apps use encryption to protect user chats and personal data.
- **Plagiarism Example:**
Copying content from a website or YouTube video and uploading it without giving credit to the original creator.

Conclusion

Software privacy and plagiarism are important issues in the digital world. Software privacy helps in protecting user information, while plagiarism laws protect original work and creativity. Both are necessary for ethical and secure use of digital technology.

